Functional Annotation Report: metXS (Q88CT3) in *Pseudomonas putida* KT2440

Gene: metXS · **Ordered locus:** PP_5097 · **UniProt:** Q88CT3 **Enzyme:** Homoserine O-succinyltransferase (HST/HTS) · **EC 2.3.1.46** **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440) **Family:** AB hydrolase superfamily, MetX family (Pfam PF00561 Abhydrolase_1; InterPro IPR008220 HAT_MetX-like, IPR029058 AB_hydrolase_fold)

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1. Summary (answer to the research question)

metXS (PP_5097) encodes a **homoserine O-succinyltransferase**, a cytoplasmic acyl-CoA–dependent acyltransferase of the MetX (α/β-hydrolase) family. Its primary function is to catalyze the **first committed step of the transsulfuration branch of L-methionine biosynthesis**:

> **succinyl-CoA** + **L-homoserine** → **CoA** + **O-succinyl-L-homoserine** (EC 2.3.1.46)

Critically, although MetX enzymes are historically assumed to be *acetyl*-transferases, the substrate specificity of this protein was **experimentally determined to be succinyl-CoA**, which is why UniProt names it *metXS* (the "S" = succinyl) rather than metX/metXA (Bastard et al. 2017, P 28581482).

The enzyme acts in the cytoplasm and its activity in *Pseudomonas* is modulated by the partner protein MetW and feedback-inhibited by S-adenosylhomocysteine.

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2. Primary function: reaction catalyzed and substrate specificity

Reaction. The enzyme transfers the succinyl moiety from succinyl-CoA onto the γ-hydroxyl of L-homoserine, producing **O-succinyl-L-homoserine** and free CoA. This is the entry reaction that diverts L-homoserine (a branch-point intermediate of the aspartate pathway) away from threonine/ isoleucine biosynthesis and into methionine biosynthesis.

Substrate specificity — the central annotation issue. MetA and MetX are two phylogenetically unrelated families of acyl-L-homoserine transferases. Members are "prone to incorrect annotation because MetX and MetA enzymes are assumed to always use acetyl-CoA and succinyl-CoA, respectively" (Bastard et al. 2017, P 28581482).

By assaying **100 enzymes** and combining the results with structural active-site classification, that study showed >60% of the ~10,000 database sequences in these families are mis-annotated, and it explicitly resolved the acyl-donor preference for individual proteins. On this basis *P. putida* PP_5097 is classified as a genuine **succinyl**transferase (metXS, EC 2.3.1.46) — a determination carried in UniProt with experimental evidence (ECO:0000269| PubMed:28581482). The study further inferred that acetyl-CoA was the ancestral substrate and that exclusive succinyl-CoA usage arose in more recently derived bacteria.

The related MetX literature confirms that acyl-donor and acyl-acceptor specificity (succinyl vs acetyl; homoserine vs serine) is dictated by residues lining the substrate-binding pocket, not by the invariant catalytic machinery. In *M. tuberculosis* homoserine acetyltransferase, single substitutions reprogrammed specificity — e.g. Leu322→Arg switched the enzyme toward **succinylation**, and Ala59Gly/ Gly62Pro created a serine-acetyltransferase-like activity (Maurya et al. 2021, P 34090964).

This establishes the molecular logic by which a MetXS such as PP_5097 achieves succinyl selectivity.

3. Catalytic mechanism and structure

The MetX/homoserine-transacylase fold is well characterized by crystallography of family members (*Haemophilus influenzae*, P 16313180; *Leptospira interrogans*, P 17927957):

- **Two-domain architecture:** a core **α/β-hydrolase domain** carrying the catalytic site, plus a **helical "lid" domain**.
- **Catalytic triad Ser–Asp–His** (e.g. Ser143/Asp304/His337 in *H. influenzae*), housed at the end of a **deep tunnel** formed at the domain interface.
- **Double-displacement (ping-pong) mechanism:** the nucleophilic serine attacks the acyl-CoA to form an **acyl(succinyl)-enzyme intermediate** with release of CoA; the succinyl group is then transferred to the hydroxyl of L-homoserine. The tunnel geometry favors transfer to the amino-acid acceptor (transferase) over water (hydrolysis) (P 16313180).

Direct sequence/annotation evidence for Q88CT3 (UniProt review, 379 aa, 41.7 kDa, homodimer): the protein carries an **AB-hydrolase domain spanning residues 51–360** and an annotated catalytic triad at **Ser157 – Asp323 – His356** (UniProt ACT_SITE 157/323/356), plus substrate-binding residues **Arg227** and **Asp357** (BINDING). Inspection of the sequence confirms these identities, and **Ser157 sits in the canonical nucleophile-elbow motif G-x-S-x-G** (local sequence ...GG**S**LG...), the hallmark of the α/β-hydrolase superfamily. Thus Q88CT3 uses the same conserved Ser–Asp–His triad and acyl-enzyme chemistry described above, with Ser157 as the catalytic nucleophile, His356 as the general base, Asp323 orienting the histidine, and Arg227 in the substrate/oxyanion-binding region — a succinyl-tuned acyl pocket.

4. Pathway context

metXS catalyzes the **first, committed step of the succinylation route to methionine**, producing the activated intermediate O-succinyl-L-homoserine. The downstream fate of this intermediate is organism-specific, and *P. putida* has been studied directly:

Primary route in *P. putida* — direct sulfhydrylation. Vermeij & Kertesz (1999, 10482527) showed that *P. putida* (strain S-313), like *P. aeruginosa*, uses **direct sulfhydrylation of O-succinylhomoserine** for methionine synthesis. In this route the succinyl intermediate reacts directly with free sulfide (H_2S) to give **L-homocysteine**, catalyzed by an O-succinylhomoserine sulfhydrylase (MetZ/MetY-type; the *metZ* gene was identified in *P. aeruginosa*, Foglino et al. 1995, 7704274). The transsulfuration enzymes (cystathionine γ-synthase and β-lyase) were expressed at low levels and only strongly upregulated when cysteine was the sole sulfur source — i.e. transsulfuration is a secondary, cysteine-inducible branch in this organism. Notably, *Pseudomonas* is unusual in performing direct sulfhydrylation on the **O-succinyl** (rather than O-acetyl) homoserine substrate (7704274).

Overall pathway topology:

1. **L-homoserine** + **succinyl-CoA → O-succinyl-L-homoserine** *(metXS / PP_5097 — this enzyme)*
2. O-succinyl-L-homoserine + sulfide → **L-homocysteine** (direct sulfhydrylation, MetZ-type; *primary route in P. putida*) — or, secondarily, O-succinyl-L-homoserine + L-cysteine → L-cystathionine (cystathionine γ-synthase) → L-homocysteine (cystathionine β-lyase) *(cysteine-inducible transsulfuration branch)*
3. L-homocysteine → **L-methionine** (methionine synthase, MetE/MetH)

Because O-succinyl-L-homoserine (rather than O-acetyl-L-homoserine) is produced, PP_5097 places *P. putida* in the **succinyl** branch typical of many Gram-negative bacteria (analogous in outcome to *E. coli* MetA, but achieved by a structurally unrelated MetX-family enzyme — a textbook case of **parallel/convergent evolution of isofunctional enzymes**, 28581482).

5. Regulation

The succinyltransferase step is a **regulated control point**. In the closely related *P. aeruginosa*, the adjacent gene **MetW** encodes a protein that physically interacts with and **activates the HST activity of MetX**; the MetW–MetX complex (but not MetX alone) is subject to **feedback inhibition by S-adenosyl-L-homocysteine (SAH)** via a glycine-rich SAM/SAH-binding motif in MetW (Gly24 required) (Hasebe 2021, P 33604638). This couples methionine-pathway flux to the cellular methylation cycle. The same metW–metX genomic arrangement and regulatory logic are the most likely physiological control on PP_5097 in *P. putida* (inference from the same genus).

6. Localization

The gene product functions in the **cytoplasm** (UniProt SUBCELLULAR LOCATION: Cytoplasm; SUBUNIT: homodimer). This is consistent with (i) its role as a soluble intermediary-metabolism enzyme acting on cytosolic substrates (L-homoserine, succinyl-CoA), (ii) the absence of a signal peptide or transmembrane segments in MetX-family proteins, and (iii) the cytosolic localization annotated for MetX/MetA-family homoserine transacylases generally. No evidence indicates secretion or membrane association.

6b. Physiological requirement in *P. putida* KT2440

A genome-wide transposon screen of *P. putida* KT2440 on glucose minimal medium recovered **methionine auxotrophs**, confirming that de novo methionine biosynthesis is functionally required (conditionally essential) when methionine is not supplied (Molina-Henares et al. 2010, P 20158506). As the enzyme catalyzing the committed entry step, metXS/PP_5097 is expected to be conditionally essential under such conditions (methionine-free minimal medium), while dispensable when methionine is available in the environment.

7. Supported and refuted hypotheses

Supported - PP_5097 is a homoserine O-**succinyl**transferase (EC 2.3.1.46), not an acetyltransferase — experimentally validated (P 28581482). - It catalyzes the first committed step of methionine biosynthesis (succinyl branch). - It uses an α/β-hydrolase fold with a Ser–Asp–His triad and double-displacement chemistry (family structural evidence). - Its activity is regulated by MetW and by SAH feedback in *Pseudomonas* (P 33604638). - It functions in the cytoplasm. - The product O-succinyl-L-homoserine is channeled **primarily via direct sulfhydrylation** to homocysteine in *P. putida* (P 10482527), with transsulfuration as a secondary, cysteine-inducible branch.

Refuted / corrected - The default database assumption that "MetX = acetyl-CoA" is **incorrect** for this protein; the metXS designation and EC 2.3.1.46 explicitly correct it. - The initial framing that the succinyl intermediate feeds mainly the *cystathionine transsulfuration* route is **corrected**: in *P. putida* the dominant downstream route is direct sulfhydrylation to homocysteine.

8. Evidence quality and limitations

- The **substrate-specificity** assignment rests on a high-quality, targeted enzymology study covering 100 family members plus structural modeling (P 28581482) — strong, but PP_5097's kinetic constants are not individually reported here.
- **Mechanism/structure** is inferred from crystal structures of MetX-family homologs (P 16313180 P 17927957) and specificity-mutation studies

(P 34090964), not from a PP_5097 structure.

- **Regulation** (MetW, SAH feedback) is demonstrated in *P. aeruginosa* (P 33604638); direct *P. putida* confirmation would strengthen the claim.
- **Localization** is by strong bioinformatic/functional inference rather than direct measurement.

Future directions: determine PP_5097 kinetics (k_cat/K_m for succinyl-CoA vs acetyl-CoA), solve its structure to map the succinyl-specificity residues, and test MetW dependence and SAH feedback directly in *P. putida* KT2440.

Key references

- Bastard et al. 2017, *Nat. Chem. Biol.* — Parallel evolution of non-homologous isofunctional enzymes in methionine biosynthesis (P 28581482). *Defining reference for the metXS/EC 2.3.1.46 assignment.*
- Hasebe 2021 — MetW regulates the enzymatic activity of MetX in *Pseudomonas* (P 33604638).
- Maurya et al. 2021 — Key residues in M. tuberculosis HAT and the evolution of the MetX family (HAT/SAT/HST) (P 34090964).
- Mirza et al. 2005 — Crystal structure of homoserine transacetylase from *H. influenzae* (P 16313180).
- Wang et al. 2007 — Crystal structure of homoserine O-acetyltransferase from *L. interrogans* (P 17927957).

- Molina-Henares et al. 2010 — Conditionally essential genes for growth of *P. putida* KT2440 on minimal medium; methionine auxotrophs recovered (P 20158506).
- Vermeij & Kertesz 1999 — Pathways of assimilative sulfur metabolism in *P. putida*; direct sulfhydrylation of O-succinylhomoserine is the primary methionine route (P 10482527).
- Foglino et al. 1995 — Direct sulfhydrylation pathway for methionine biosynthesis in *P. aeruginosa*; identification of *metZ* (P 7704274).
- UniProt Q88CT3 (METXS_PSEPK) — reviewed entry: 379 aa homodimer, cytoplasm, ACT_SITE Ser157/Asp323/His356, EC 2.3.1.46.